

Task 1 — Launch a Basic Web Server

Steps

1. Launch an EC2 instance.
2. Install a web server.
3. Create a simple HTML page.
4. Configure HTTP access in the security group.
5. Open the instance's public address in a browser.

Example:

```
<!DOCTYPE html>
<html>
<head>
  <title>Cloud Lab</title>
</head>
<body>
  <h1>Welcome to My Cloud Website</h1>
  <p>This website is hosted on AWS EC2.</p>
</body>
</html>
```

Public URL

Record your actual URL:

- <http://YOUR-PUBLIC-IP/>

Task 2 — Create an Auto Scaling Group

Recommended configuration

Minimum capacity: 1

Desired capacity: 1

Maximum capacity: 3

Steps

1. Create a launch template.

2. Select the appropriate AMI.
3. Select the instance type eligible for your lab/account.
4. Configure the security group.
5. Add your web-server startup configuration if needed.
6. Open EC2 → Auto Scaling Groups.
7. Select Create Auto Scaling group.
8. Select your launch template.
9. Select your VPC and subnets.
10. Set:
 - a. Minimum: 1
 - b. Desired: 1
 - c. Maximum: 3
11. Configure a target-tracking or other suitable scaling policy based on CPU utilization.

Task 3 — Simulate Increased Traffic

Possible tools include:

- k6
- ApacheBench
- Locust
- [Loader.io](https://loader.io/)

Monitor:

- CloudWatch → EC2 → CPUUtilization

Task 4 — Scale Down

1. Stop the load test.
2. Allow CPU utilization to decrease.
3. Wait for the configured scale-in conditions.
4. Open:
 - EC2 → Auto Scaling Groups → Activity
5. Check the activity history.

6. Verify that the group eventually returns toward its configured minimum.

Task 5 — Real-World Auto Scaling Scenarios

1. Zomato — Dinner-Time Traffic

Traffic can increase significantly around lunch and dinner. Auto Scaling can add application servers during busy periods and reduce them when traffic falls.

2. Flipkart — Big Sale

During major sales, millions of users may access products simultaneously. Auto Scaling helps provide additional computing capacity during the traffic spike.

3. BookMyShow — IPL/Movie Ticket Release

When tickets become available, many users can attempt to purchase them simultaneously. Auto Scaling helps increase application capacity and maintain responsiveness.